

Nutritional Literacy Intervention of 1st and 2nd Graders within DCPS

Is there a difference in nutritional literacy scores between the “control” schools versus intervention schools for grades 1 and 2?

H_1 = There is evidence of nutritional literacy intervention impact between the intervention and comparison schools among 1st and 2nd graders

Comparison School Score Change Data

- o 1st graders (n=22, $\bar{x}$ =5.555)
- o 2nd graders (n=25, $\bar{x}$ =6.427)
- o Overall Cohen's D Effect Size: 20.847
- o p=.551.
- o Since n=46, the df is 45(46-1).

Intervention School Score Change Data

- o 1st graders (n=31, $\bar{x}$ =8.234)
- o 2nd graders (n=22, $\bar{x}$ =21.212)
- o Overall Cohen's D Effect Size: 28.081
- o p=.006.
- o Since n=52, the df is 51(52-1).

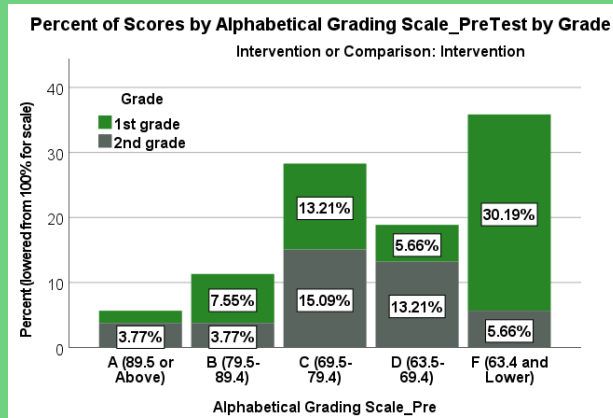
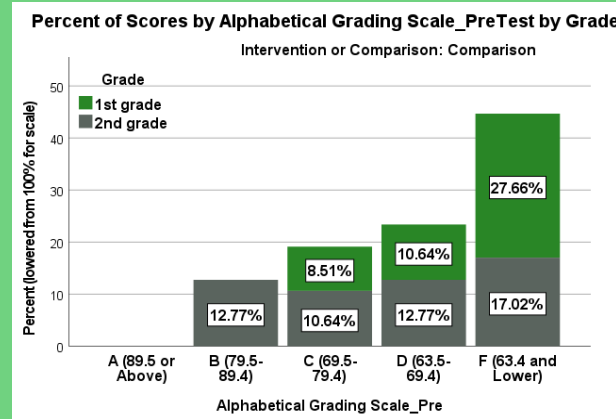
Analysis: An independent t-test of the separate, unrelated variables between the school type, grade levels and scores change between the pre and post-tests was conducted. The students chosen for this project were limited only to those who took both the pre-test and post-test for fairer, more reliable analysis of nutritional literacy impact.

Summary: In the pretest, the students in both the intervention school (54.72%) and comparison schools (68.09%) had an over 50% nutritional literacy score of a 63.5 and lower (alphabetical score of D to F). What specifically caused the improvement in score for the comparison school will require further investigation, but in the post test, we see an increase in both schools in nutritional literacy. The score change itself appears to be statistically notable in the intervention school as most of the students now scored above 63.5 (alphabetical score of D) at 73.91%.

The mean score change difference of the intervention school was 12.96 while the comparison school was .871. While the level of significance for the comparison school was greater than .05 at p=.551 listing no statistical significance, the intervention school showed a high level of statistical significance at p=.006. This is evidence to accept our hypothesis of nutritional literacy intervention impact within the intervention school.

Summer 2021

Pre-Test Scores



Post-Test Scores

